JEE Mathematics: Quick Review with Grading and Practice Map - Hyperbola (Batch XIII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (37 questions; a question that appears in two of these is counted once): High = 3 or more questions, Medium = 1 to 2 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Hyperbola: Introduction

Question 01

What is the condition for general equation of 2nd degree curve to represent a hyperbola?

Model Answer

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0, \quad h^2 > ab$$

If also $a + b = 0$, it is a rectangular hyperbola. ($\Delta = 0$ gives a pair of lines.)

Ex: $x^2 - 3y^2 - 4x - 6y - 11 = 0$: $h = 0 > ab = -3$, $\Delta \neq 0 \Rightarrow$ hyperbola.

Grade - Low

Practice map, main concept: Sheet: J-I 15; J-II 75, 95

Also used in: none

Question 02

What is the definition of hyperbola?

Model Answer

Locus of a point whose distances from two fixed points (foci) have a constant difference: $|PS_1 - PS_2| = 2a < S_1S_2$. Also $PS = e \cdot PM$ with $e > 1$.

$2a = S_1S_2$ gives two rays (outside the segment S_1S_2); $2a > S_1S_2$ gives no locus.

Grade - Low

Practice map, main concept: Sheet: C-II 63

Also used in: none

Question 03

What is the equation of standard hyperbola? And what is the relation between a and b?

Model Answer

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \qquad b^2 = a^2(e^2 - 1) = c^2 - a^2$$

No order between a and b is needed ($a < b$, $a = b$, $a > b$ all allowed); a is under the positive term.

Grade - Medium

Practice map, main concept: Sheet: J-II 11, 63; C-I 20 | DPP: 8(7) | Question Bank: 116

Also used in: Sheet: J-I 15; J-II 5, 75

Question 04

Define transverse axis / conjugate axis, foci, vertices, centre, principal axes for a hyperbola.

Model Answer

Transverse axis (TA) = line through the foci, length $2a$, ends = vertices $(\pm a, 0)$. Conjugate axis (CA) = perpendicular bisector of S_1S_2, length $2b$, ends $(0, \pm b)$.

Centre = intersection of the axes; TA and CA together = principal axes; foci $(\pm ae, 0)$, focal length $= 2ae$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 05

Define focal distance / focal radius, focal chord, double ordinate, latus rectum for a hyperbola.

Model Answer

Focal distance = distance of a point from a focus; focal chord = chord through a focus; double ordinate = chord perpendicular to TA; latus rectum = double ordinate through a focus.

Ex: double ordinate AB with $\triangle AOB$ equilateral: $(a\sec\theta, \pm b\tan\theta)$ with $\frac{b}{a}\sin\theta = \frac{1}{\sqrt{3}}$ needs $\frac{b}{a} > \frac{1}{\sqrt{3}}$, i.e. $e > \frac{2}{\sqrt{3}}$.

Grade - Low

Practice map, main concept: DPP: 8(19)

Also used in: Sheet: J-II 13

Question 06

What is the formula of eccentricity?

Model Answer

$$e = \frac{c}{a} = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{1 + \frac{(CA)^2}{(TA)^2}} > 1$$

Ex: $\frac{x^2}{16} - \frac{y^2}{9} = 1$: $e = \sqrt{1 + \frac{9}{16}} = \frac{5}{4}$.

Grade - High

Practice map, main concept: Sheet: J-II 82, 91; J-M 31, 44, 47-48, 50, 62; C-II 44 | DPP: 8(4) | Question Bank: 114, 118-119

Also used in: Sheet: J-I 11-12, 14, 31; J-II 6; J-M 14, 34, 46, 49; J-A 27, 32 | DPP: 8(3, 9, 13, 17-19); 9(3, 15, 18); 10(6) | Question Bank: 111, 121, 123-125, 128

Question 07

Write expression for b in terms of a and e.

Model Answer

$$b^2 = a^2(e^2 - 1), \qquad b = a\sqrt{e^2 - 1}$$

Grade - Low

Practice map, main concept: none

Also used in: none

Question 08
What are the coordinates of ends of latus rectum and length of latus rectum for standard hyperbola?

Model Answer

$$\left(\pm ae,\ \pm\frac{b^2}{a}\right), \qquad \ell_{LR} = \frac{2b^2}{a} = \frac{(CA)^2}{TA} = 2a(e^2-1)$$

Ex: $\frac{x^2}{144} - \frac{y^2}{25} = 1$: ends $\left(\pm 13, \pm\frac{25}{12}\right)$, $\ell_{LR} = \frac{25}{6}$.

Grade - High
Practice map, main concept: Sheet: J-II 5, 57; J-M 14, 49 | DPP: 8(9, 13, 17) | Question Bank: 123, 125, 128
Also used in: Sheet: J-I 16; J-M 31, 61; J-A 27; C-II 39, 41 | DPP: 9(4, 14) | Question Bank: 111-112, 115

Question 09
What is the relation between length of LR, e and distance between focus and corresponding directrix for a hyperbola?

Model Answer

$$\ell_{LR} = 2e \times \left(ae - \frac{a}{e}\right) = 2e \times (\text{focus–directrix distance})$$

Ex: focus $(1, 1)$, directrix $2x + y = 1$, $e = \sqrt{3}$: distance $= \frac{2}{\sqrt{5}}$, so $\ell_{LR} = \frac{4\sqrt{3}}{\sqrt{5}}$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 4

Question 10
How to find focal distance of a point on a hyperbola?

Model Answer

$$PS_1 = |ex_1 - a|, \qquad PS_2 = |ex_1 + a|$$

Use $PS = e \cdot PM$. Product $PS_1 \cdot PS_2 = e^2x_1^2 - a^2 = CP^2 - a^2 + b^2$; on the right branch $PS_2 - PS_1 = 2a$.
Ex: $(5, \frac{9}{4})$ on $\frac{x^2}{16} - \frac{y^2}{9} = 1$: $PS_1 = \frac{25}{4} - 4 = \frac{9}{4}$, $PS_2 = \frac{41}{4}$.

Grade - High
Practice map, main concept: Sheet: J-II 56; J-M 46, 61, 63 | Question Bank: 112, 115, 121, 126
Also used in: Sheet: J-II 58

Question 11
How to check whether two hyperbolas are similar?

Model Answer
Two hyperbolas are similar if they have the same eccentricity.

Grade - Low
Practice map, main concept: none
Also used in: none

Auxiliary Circle and Eccentric Angle

Question 12

Define auxiliary circle for a hyperbola.

Model Answer

The circle with centre C and the transverse axis as diameter.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 13

What is the equation of auxiliary circle for standard hyperbola?

Model Answer

$$x^2 + y^2 = a^2$$

It is also the circle of least radius centred at C that touches the hyperbola (at the vertices).

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-M 60 | Question Bank: 117

Question 14

What are parametric coordinates of a point on standard hyperbola?

Model Answer

$$P(\theta) = (a\sec\theta,\ b\tan\theta), \quad \theta \in [0, 2\pi),\ \theta \neq \frac{\pi}{2}, \frac{3\pi}{2}$$

Other forms: $\left(\frac{a}{2}\left(t + \frac{1}{t}\right), \frac{b}{2}\left(t - \frac{1}{t}\right)\right)$; eliminating a parameter from two lines often gives a hyperbola ($\sec^2\theta - \tan^2\theta = 1$).

Grade - Medium

Practice map, main concept: Sheet: J-II 48, 74 | DPP: 8(3, 10, 14)

Also used in: Sheet: J-M 45; C-II 45, 53 | Question Bank: 113, 120

Question 15

What is eccentric angle of a point on hyperbola?

Model Answer

θ in $P(a\sec\theta, b\tan\theta)$. Draw $PN \perp$ TA, tangent from $N(a\sec\theta, 0)$ to the auxiliary circle touches it at $Q(a\cos\theta, a\sin\theta)$; $\theta = \angle QCN$.

Quadrants: θ in 1st $\rightarrow$ P in 1st, 2nd $\rightarrow$ 3rd, 3rd $\rightarrow$ 2nd, 4th $\rightarrow$ 4th.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 16
Define corresponding points.

Model Answer
$P(a\sec\theta, b\tan\theta)$ on the hyperbola and $Q(a\cos\theta, a\sin\theta)$ on the auxiliary circle (same θ) are corresponding points.

Grade - Low
Practice map, main concept: none
Also used in: none

Standard Hyperbolas

Question 17
What does 'everything' mean for a hyperbola?

Model Answer
Centre, vertices, ends of conjugate axis, e, foci, feet of directrices, directrices, ends of latus rectum, length of latus rectum and auxiliary circle.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 18
Find everything for standard hyperbola.

Model Answer

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Centre $(0,0)$; vertices $(\pm a, 0)$; CA ends $(0, \pm b)$; foci $(\pm ae, 0)$; directrices $x = \pm\frac{a}{e}$; LR ends $\left(\pm ae, \pm\frac{b^2}{a}\right)$; $\ell_{LR} = \frac{2b^2}{a}$; aux. circle $x^2 + y^2 = a^2$; $e^2 = 1 + \frac{b^2}{a^2}$.

Ex: $\frac{x^2}{144} - \frac{y^2}{25} = 1$: $e = \frac{13}{12}$, foci $(\pm 13, 0)$, directrices $x = \pm\frac{144}{13}$, $\ell_{LR} = \frac{25}{6}$.

$\frac{x^2}{\cos^2\alpha} - \frac{y^2}{\sin^2\alpha} = 1$: $a^2 + b^2 = 1$, so the foci $(\pm 1, 0)$ stay fixed as α varies.

Grade - High
Practice map, main concept: Sheet: J-I 11-12, 14; J-II 6; J-M 45, 60, 64; J-A 27 | DPP: 8(5, 15-16, 18) | Question Bank: 111, 117, 120, 124, 127
Also used in: Sheet: J-II 38, 57, 82; J-M 44; C-II 44 | DPP: 8(7); 9(10) | Question Bank: 114, 119

Question 19
Find everything for standard hyperbola (vertical).

Model Answer

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1 \quad \left(\text{i.e. } \frac{x^2}{b^2} - \frac{y^2}{a^2} = -1\right)$$

Vertices $(0, \pm a)$; CA ends $(\pm b, 0)$; foci $(0, \pm ae)$; directrices $y = \pm\frac{a}{e}$; LR ends $\left(\pm\frac{b^2}{a}, \pm ae\right)$; $e^2 = 1 + \frac{b^2}{a^2}$.

Ex: $\frac{y^2}{16} - \frac{x^2}{9} = 1$: $e = \frac{5}{4}$, foci $(0, \pm 5)$, directrices $y = \pm\frac{16}{5}$, $\ell_{LR} = \frac{9}{2}$.

Grade - Medium
Practice map, main concept: DPP: 9(18)
Also used in: Sheet: J-II 91; J-M 46; C-I 5 | Question Bank: 121

Question 20

Plot both types of standard hyperbola and mark everything for them.

Model Answer

Draw the axes and the asymptotes $y = \pm\frac{b}{a}x$ (diagonals of the $2a \times 2b$ rectangle); the two branches pass through the vertices and approach the asymptotes; foci lie outside the vertices at ae, directrices between centre and vertices at $\frac{a}{e}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 21

Find everything for hyperbola if equation is given in PS = e PM form.

Model Answer

Read focus S, directrix and $e > 1$. TA = perpendicular from S to the directrix; $ae - \frac{a}{e}$ = focus–directrix distance; centre on the TA at distance ae from S, on the directrix side.

Ex: focus $(1, 1)$, directrix $2x + y = 1$, $e = \sqrt{3}$: $5\left[(x-1)^2 + (y-1)^2\right] = 3(2x + y - 1)^2 \Rightarrow 7x^2 + 12xy - 2y^2 - 2x + 4y - 7 = 0$.

Grade - Low

Practice map, main concept: Sheet: J-II 4; C-II 46 | DPP: 8(11)

Also used in: none

Question 22

How is the equation of a shifted hyperbola represented?

Model Answer

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

Centre (h, k), axes parallel to the coordinate axes; put $X = x - h$, $Y = y - k$ to get the standard form.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 23

How to observe whether equation of a 2nd degree curve represents shifted hyperbola?

Model Answer

No xy term and the coefficients of x^2, y^2 have opposite signs; complete the squares and check that the right side is non-zero.

Right side = 0 gives a pair of lines (the asymptotes); a negative right side gives the conjugate (vertical) form.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 24

Find everything for shifted hyperbola.

Model Answer

Find everything in X, Y, then put back $x = X + h$, $y = Y + k$.

Ex: $16x^2 - 9y^2 + 32x + 36y - 164 = 0 \Rightarrow \frac{(x+1)^2}{9} - \frac{(y-2)^2}{16} = 1$: centre $(-1, 2)$, $e = \frac{5}{3}$, foci $(4, 2), (-6, 2)$, $\ell_{LR} = \frac{32}{3}$, directrices $x = -1 \pm \frac{9}{5}$.

Grade - Low

Practice map, main concept: Sheet: J-I 16; J-II 7; C-II 43 | DPP: 8(1, 6, 12, 20); 9(14-15)

Also used in: DPP: 9(13)

Question 25

How to find equation of shifted hyperbola if some details about it are given?

Model Answer

Centre = mid-point of the vertices (or foci); direction of TA from them; then a = half the vertex distance, ae = half the focal distance, $b^2 = a^2(e^2 - 1)$.

Ex: vertices $(0, 0), (10, 0)$, a focus $(18, 0)$: centre $(5, 0)$, $a = 5$, $ae = 13$, $b^2 = 144$: $\frac{(x-5)^2}{25} - \frac{y^2}{144} = 1$.

Grade - Medium

Practice map, main concept: DPP: 8(8) | Question Bank: 122

Also used in: Sheet: J-M 62 | Question Bank: 118

Question 26

What is the locus of centre of a variable circle which touches two fixed circles externally?

Model Answer

$PC_1 = r + r_1$, $PC_2 = r + r_2 \Rightarrow |PC_1 - PC_2| = |r_1 - r_2|$: one branch of a hyperbola with foci C_1, C_2 and $2a = |r_1 - r_2|$ (a line if $r_1 = r_2$).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 27

Find everything for hyperbola if equation is given in $|PS_1 - PS_2| = 2a$ form.

Model Answer

Foci = the two fixed points; centre = their mid-point; $2ae = S_1S_2$; $e = \frac{S_1S_2}{2a}$; TA along S_1S_2. Only $PS_1 - PS_2 = +2a$ (not $\pm$) gives one branch.

Ex: $\sqrt{(x-1)^2 + (y-2)^2} - \sqrt{(x-5)^2 + (y-5)^2} = 3$: $S_1S_2 = 5$, $2a = 3$, $e = \frac{5}{3}$; conjugate $e' = \frac{5}{4}$.

Grade - Low

Practice map, main concept: Sheet: J-I 13; C-II 65

Also used in: none

Alternate Definition of Hyperbola

Question 28
Derive alternate definition of hyperbola.

Model Answer

$$\frac{p_2^2}{a^2} - \frac{p_1^2}{b^2} = 1$$

p_1 = distance of P from the transverse axis ($|y|$), p_2 = distance from the conjugate axis ($|x|$); so $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is this statement for the axes $y = 0$, $x = 0$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 29
Find everything for hyperbola if equation is given in alternate definition form.

Model Answer
Write each bracket as a true distance (divide by $\sqrt{l^2 + m^2}$). The line under a^2 (positive term) is the CONJUGATE axis, the other is the transverse axis; centre = their intersection.
Ex: $\frac{(x+y-2)^2}{18} - \frac{(x-y)^2}{8} = 1 \Rightarrow \frac{p_2^2}{9} - \frac{p_1^2}{4} = 1$: $a = 3, b = 2$, $e = \frac{\sqrt{13}}{3}$, $\ell_{LR} = \frac{8}{3}$, TA $x = y$, centre $(1, 1)$.

Grade - Low
Practice map, main concept: none
Also used in: none

Conjugate Hyperbola

Question 30
Define conjugate hyperbola.

Model Answer
The hyperbola whose transverse and conjugate axes are respectively the conjugate and transverse axes of the given one.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 31
How to find equation of conjugate hyperbola for a standard hyperbola?

Model Answer

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \longrightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$$

Change the sign of the constant (equivalently of the x^2, y^2 terms). Foci $(0, \pm be')$, $\ell_{LR} = \frac{2a^2}{b}$.

Grade - Low
Practice map, main concept: DPP: 9(2)
Also used in: DPP: 8(2); 9(18)

Question 32
What is the relation between eccentricities of a hyperbola and its conjugate hyperbola?

Model Answer

$$\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$$

The four foci are concyclic (circle $x^2 + y^2 = a^2 + b^2$) and form a square.
Ex: $\frac{x^2}{4} - \frac{y^2}{12} = 1$: $e_1 = 2 \Rightarrow e_2 = \frac{2}{\sqrt{3}}$.

Grade - Low
Practice map, main concept: DPP: 8(2)
Also used in: Sheet: J-I 13 | DPP: 9(2)

Rectangular Hyperbola

Question 33
What is the condition for a hyperbola to be rectangular?

Model Answer
$a = b \iff e = \sqrt{2} \iff$ asymptotes perpendicular
Standard form $x^2 - y^2 = a^2$; in the general equation, $a + b = 0$ (coefficients of x^2, y^2).

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-II 81; J-M 30; C-I 4

Question 34
What is the relation between lengths of latus rectum, TA and CA for a rectangular hyperbola?

Model Answer

$$\ell_{LR} = \ell_{TA} = \ell_{CA} = 2a$$

Grade - Low
Practice map, main concept: none
Also used in: none

Position of a Point w.r.t. Hyperbola

Question 35
Which region is called inside / outside w.r.t. a hyperbola?

Model Answer
Inside = the regions containing the foci (enclosed by each branch); outside = the region containing the centre.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 36
Write S, S_1 and T for standard hyperbola.

Model Answer

$$S = \frac{x^2}{a^2} - \frac{y^2}{b^2} - 1, \quad S_1 = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1, \quad T = \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$$

Grade - Low

Practice map, main concept: none

Also used in: none

Question 37
How to find relative position of a point w.r.t. a hyperbola?

Model Answer
$S_1 > 0$: inside; $S_1 = 0$: on; $S_1 < 0$: outside (the centre gives $S_1 = -1 < 0$). Write S with the constant as -1 on the left first.
Ex: $(2, 1)$ and $\frac{x^2}{4} - y^2 = 1$: $S_1 = 1 - 1 - 1 = -1 < 0$, outside.

Grade - Low

Practice map, main concept: none

Also used in: none

Chord Joining Two Points

Question 38
Write expression of equation of chord joining two points $P(\alpha)$ and $Q(\beta)$.

Model Answer

$$\frac{x}{a}\cos\frac{\alpha-\beta}{2} - \frac{y}{b}\sin\frac{\alpha+\beta}{2} = \cos\frac{\alpha+\beta}{2}$$

Through $(d, 0)$: $\tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{a-d}{a+d}$; focal chord through $(ae, 0)$: $\tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{1-e}{1+e}$.

Grade - Low

Practice map, main concept: Sheet: J-II 13

Also used in: Sheet: C-II 48

Line and Hyperbola

Question 39
What is condition of tangency for standard hyperbola? And how is it used?

Model Answer
$y = mx + c$ touches $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \iff c^2 = a^2m^2 - b^2$

Tangent of slope m: $y = mx \pm \sqrt{a^2m^2 - b^2}$ (exists only for $|m| > \frac{b}{a}$). Line $lx + my + n = 0$: $a^2l^2 - b^2m^2 = n^2$.
Ex: $\frac{x^2}{16} - \frac{y^2}{9} = 1$, slope 2: $y = 2x \pm \sqrt{55}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 29; J-II 38, 62, 94; J-A 22 | DPP: 9(5, 12); 10(14)

Also used in: Sheet: J-II 16, 34-35; J-M 63 | DPP: 9(7) | Question Bank: 126

Question 40

How to apply condition of tangency for different types of standard hyperbolas?

Model Answer

Write the curve as $\frac{x^2}{A} - \frac{y^2}{B} = 1$ with signed A, B and use $c^2 = Am^2 - B$.

Ex: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$ is $\frac{x^2}{-a^2} - \frac{y^2}{-b^2} = 1$: $c^2 = b^2 - a^2m^2$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-M 33

Director Circle

Question 41

What is equation of director circle for a standard hyperbola?

Model Answer

$$x^2 + y^2 = a^2 - b^2$$

Real only if $a > b$ ($e < \sqrt{2}$); point circle (centre) if $a = b$; no real point if $a < b$.

Ex: $\frac{x^2}{16} - \frac{y^2}{9} = 1$: $x^2 + y^2 = 7$.

Grade - Low

Practice map, main concept: Sheet: J-II 90; C-II 60 | DPP: 9(3)

Also used in: DPP: 8(20)

Question 42

Derive equation of director circle of standard hyperbola.

Model Answer

Tangent $y = mx \pm \sqrt{a^2m^2 - b^2}$ through (h, k): $(h^2 - a^2)m^2 - 2hkm + k^2 + b^2 = 0$. Perpendicular: $m_1m_2 = \frac{k^2+b^2}{h^2-a^2} = -1 \Rightarrow h^2 + k^2 = a^2 - b^2$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 43

How to find locus of a variable external point P for which condition on slope of tangents drawn from P to hyperbola is given?

Model Answer

Use $m_1 + m_2 = \frac{2hk}{h^2-a^2}$, $m_1m_2 = \frac{k^2+b^2}{h^2-a^2}$ in the given condition, then replace (h, k) by (x, y).

Ex: $3x^2 - 2y^2 = 6$, $\tan\theta_1 \tan\theta_2 = 2$: $\frac{k^2+3}{h^2-2} = 2 \Rightarrow y^2 = 2x^2 - 7$.

Grade - Medium

Practice map, main concept: Sheet: J-II 16; J-M 33

Also used in: none

Tangent & Normal

Question 44
How to find equation of tangent for a hyperbola if point of tangency is given?

Model Answer

$$T = 0: \quad \frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

Ex: at $(5, \frac{9}{4})$ on $\frac{x^2}{16} - \frac{y^2}{9} = 1$: $\frac{5x}{16} - \frac{y}{4} = 1$, i.e. $5x - 4y = 16$.
Slope at an end of LR $= \pm e$; tangent meets TA at T with $CT \cdot CN = a^2$ (N = foot of ordinate).

Grade - Medium
Practice map, main concept: Sheet: J-A 16, 32; C-II 39 | DPP: 9(1, 4)
Also used in: Sheet: C-II 42, 49

Question 45
How to find equation of tangent for a hyperbola if any other condition other than point of tangency is given?

Model Answer
Take the slope form $y = mx \pm \sqrt{a^2m^2 - b^2}$ (slope / parallel / perpendicular / through a point) or the parametric form $\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$ (intercepts, loci) and apply the condition.
Ex: $x^2 - 4y^2 = 36$, perpendicular to $x - y + 4 = 0$: $m = -1$, $c^2 = 36 - 9$: $y = -x \pm 3\sqrt{3}$.
Foot of perpendicular from the centre on a tangent: locus $(x^2 + y^2)^2 = a^2x^2 - b^2y^2$.

Grade - Medium
Practice map, main concept: Sheet: J-II 34; J-M 34 | DPP: 9(7)
Also used in: Sheet: J-II 62; J-A 3

Question 46
What is the equation of tangent at a point, whose coordinates are given in parametric form, on standard hyperbola?

Model Answer

$$\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$$

Perpendicular from the centre: $p = \frac{ab}{\sqrt{b^2\sec^2\theta + a^2\tan^2\theta}}$.

Grade - Low
Practice map, main concept: Sheet: J-II 58; C-II 42, 45
Also used in: Sheet: C-II 51

Question 47
How to find equation of tangent to standard hyperbola passing through an external point?

Model Answer
Put (h, k) in $y = mx \pm \sqrt{a^2m^2 - b^2}$: $(h^2 - a^2)m^2 - 2hkm + k^2 + b^2 = 0$, solve for m (if $h^2 = a^2$, one tangent is vertical).
Ex: $\frac{x^2}{36} - \frac{y^2}{9} = 1$ from $(0, 4)$: $16 = 36m^2 - 9$, $m = \pm\frac{5}{6}$: $5x - 6y + 24 = 0$, $5x + 6y - 24 = 0$.

Grade - Low
Practice map, main concept: Sheet: J-II 35; C-II 40 | DPP: 9(13); 10(13)
Also used in: Sheet: J-I 30

Question 48

How to find point of tangency if equation of tangent is given?

Model Answer

$$y = mx + c \text{ touches at } \left(-\frac{a^2 m}{c},\ -\frac{b^2}{c}\right)$$

Or compare the line with $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$.

Ex: $y = x + 2$ on $5x^2 - 9y^2 = 45$: $c^2 = 9 - 5 = 4$ holds, so $c = 2$; point $\left(-\frac{9}{2}, -\frac{5}{2}\right)$.

Grade - High

Practice map, main concept: Sheet: J-A 3 | DPP: 9(8-9)

Also used in: Sheet: J-M 13, 63 | Question Bank: 126

Question 49

How to find equation of normal for a standard hyperbola if foot of normal is given?

Model Answer

$$\frac{a^2 x}{x_1} + \frac{b^2 y}{y_1} = a^2 + b^2$$

Normal meets TA at $G(e^2 x_1, 0)$ and CA at $\left(0, \frac{a^2+b^2}{b^2} y_1\right)$.

Ex: normal at $(6, 3)$ meets x-axis at $(9, 0)$: $9 = 6e^2$, $e = \sqrt{\frac{3}{2}}$ $(a^2 = 18, b^2 = 9)$.

Grade - High

Practice map, main concept: Sheet: J-I 31; J-M 32; C-II 41

Also used in: Sheet: J-A 16, 32; C-II 39

Question 50

Write expression for equation of normal to standard hyperbola in terms of eccentric angle θ.

Model Answer

$$\frac{ax}{\sec\theta} + \frac{by}{\tan\theta} = a^2 + b^2 = a^2 e^2$$

Grade - Medium

Practice map, main concept: Sheet: J-II 36-37; C-II 48-49

Also used in: Sheet: J-M 15

Question 51

How to find the condition for a line to be a normal to standard hyperbola?

Model Answer

$$y = mx + c \text{ is a normal } \iff c^2 = \frac{m^2(a^2 + b^2)^2}{a^2 - b^2 m^2}$$

Compare $y = mx + c$ with the parametric normal and eliminate θ. From a point at most four normals can be drawn.

Grade - Medium

Practice map, main concept: Sheet: J-M 15 | DPP: 9(11)

Also used in: Sheet: J-II 36

Question 52

How to find point of intersection of tangents drawn at $P(\alpha)$ and $Q(\beta)$ on standard hyperbola?

Model Answer

$$\left(\frac{a\cos\frac{\alpha-\beta}{2}}{\cos\frac{\alpha+\beta}{2}},\ \frac{b\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha+\beta}{2}}\right)$$

Compare the chord of contact of (h,k) with the chord PQ. Ex: $\alpha+\beta=\frac{\pi}{2}$ gives the locus $y=b$.

Grade - Low

Practice map, main concept: DPP: 9(6)

Also used in: none

Question 53

How to find minimum distance between a point and standard hyperbola?

Model Answer

The shortest segment lies along a normal: write the normal at $P(\theta)$ through the point, find θ, then the distance.

Ex: from a focus the nearest point is the nearer vertex: least distance $= ae - a$ (e.g. $\frac{x^2}{9}-\frac{y^2}{16}=1$: $5-3=2$).

Grade - Low

Practice map, main concept: DPP: 9(10)

Also used in: none

Question 54

How to find minimum distance between a line and standard hyperbola?

Model Answer

If the line does not meet the curve, draw the tangents parallel to it ($c^2=a^2m^2-b^2$); the distance between the line and the nearer tangent is the answer, the point of contact is the nearest point.

Ex: $\frac{x^2}{4}-y^2=1$ and $y=x+1$: tangents $y=x\pm\sqrt{3}$; least distance $=\frac{\sqrt{3}-1}{\sqrt{2}}$.

Grade - Medium

Practice map, main concept: Sheet: J-M 13

Also used in: none

Common Concepts

Question 55

Write expression for equation of chord whose mid point is given for standard hyperbola.

Model Answer

$$T=S_1:\quad \frac{xx_1}{a^2}-\frac{yy_1}{b^2}=\frac{x_1^2}{a^2}-\frac{y_1^2}{b^2}$$

Ex: $x^2-y^2=9$ bisected at $(5,-3)$: $5x+3y=25-9$, i.e. $5x+3y=16$.

Grade - Low

Practice map, main concept: DPP: 10(4)

Also used in: none

Question 56
Write expression for equation of chord of contact for standard hyperbola.

Model Answer

$$T = 0: \quad \frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

Ex: from $(3, 2)$ to $x^2 - 9y^2 = 9$: $3x - 18y = 9$, i.e. $x - 6y = 3$.

Grade - **Low**
Practice map, main concept: Sheet: J-I 30
Also used in: Sheet: J-II 47; C-II 40 | DPP: 10(13)

Question 57
How to find mid point of a chord of a standard hyperbola whose equation is given?

Model Answer
Compare $lx + my = n$ with $T = S_1$: $\frac{h/a^2}{l} = \frac{-k/b^2}{m} = \frac{h^2/a^2 - k^2/b^2}{n}$.
Locus problems: write the chord as $T = S_1$ and impose the condition (passes through a point, touches a curve, fixed slope...).

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 58
How to find external point for a chord of contact for standard hyperbola if equation of chord of contact is given?

Model Answer
Compare the given line with $\frac{hx}{a^2} - \frac{ky}{b^2} = 1$.
Ex: $QR: 4x - 3y = 6$ for $\frac{x^2}{9} - \frac{y^2}{4} = 1$: $\frac{h}{9} = \frac{4}{6}$, $\frac{k}{4} = \frac{3}{6}$: $P(6, 2)$.

Grade - **Low**
Practice map, main concept: DPP: 10(1)
Also used in: none

Question 59
Write expression for combined equation of pair of tangents for standard hyperbola.

Model Answer

$$SS_1 = T^2$$

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 60
Write expression for equation of polar for standard hyperbola.

Model Answer

$$T = 0: \quad \frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

Polar of (x_1, y_1) = locus of the intersection of tangents at the ends of chords through (x_1, y_1).

Grade - **Low**
Practice map, main concept: none

Also used in: DPP: 10(3)

Question 61

What does $T = 0$ represent for a point, position relative to standard hyperbola?

Model Answer

On the curve: tangent; outside (tangents exist): chord of contact; any point: polar. Chord of contact is a part of the polar.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 62

How to find coordinates of pole of a given line?

Model Answer

$$lx + my + n = 0 \;\Rightarrow\; \text{pole}\left(-\frac{a^2 l}{n},\; \frac{b^2 m}{n}\right)$$

Compare with $\frac{hx}{a^2} - \frac{ky}{b^2} = 1$.

Grade - Low

Practice map, main concept: Sheet: J-II 47

Also used in: none

Question 63

From any variable point on a fixed line, tangents are drawn to the hyperbola; then chord of contact will pass through which point?

Model Answer

Through the pole of the fixed line. Conversely, chords through a fixed point: tangents at their ends meet on the polar of that point.

Ex: chords of $\frac{x^2}{16} - \frac{y^2}{9} = 1$ through $(2, 1)$: locus $\frac{2x}{16} - \frac{y}{9} = 1$, i.e. $9x - 8y = 72$.

Grade - Low

Practice map, main concept: DPP: 10(3)

Also used in: none

Question 64

Derive equation of diameter for standard hyperbola.

Model Answer

$$y = \frac{b^2}{a^2 m}\, x$$

Mid-points (h, k) of chords of slope m: from $T = S_1$, slope $\frac{b^2 h}{a^2 k} = m$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-I 43

Highlights

Question 65
What is the difference of focal distances of any point on hyperbola?

Model Answer

$$|PS - PS'| = 2a$$

$PS' - PS = 2a$: right branch (nearer S); $PS - PS' = 2a$: left branch.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-II 56; J-A 39; C-II 43

Question 66
What is the product of length of perpendiculars drawn from foci on any variable tangent to the hyperbola?

Model Answer

$$p_1 p_2 = b^2$$

Distance of a focus from either asymptote $= b$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 98

Question 67
What angle does portion of tangent intercepted between directrix and point of tangency subtend at focus?

Model Answer
A right angle (at the corresponding focus).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 68
What is the relation between auxiliary circle and circle drawn on any focal distance as diameter?

Model Answer
They touch each other.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 69
Where does foot of perpendicular from focus on any tangent to hyperbola lie?

Model Answer
On the auxiliary circle $x^2 + y^2 = a^2$.
Ex: $16y^2 - 9x^2 = 1$ is vertical with $a^2 = \frac{1}{16}$: locus $x^2 + y^2 = \frac{1}{16}$.

Grade - Low
Practice map, main concept: DPP: 10(5)
Also used in: none

Question 70
Explain reflection property of hyperbola.

Model Answer
Tangent and normal at P bisect the angles between the focal radii PS, PS' (tangent = internal bisector). A ray aimed at one focus is reflected from the outer surface towards the other focus.

Grade - Medium
Practice map, main concept: Sheet: J-A 39
Also used in: none

Question 71
What is the angle of intersection of an ellipse and a hyperbola if they are confocal?

Model Answer
90°: confocal ellipse and hyperbola cut orthogonally. E.g. $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and $\frac{x^2}{a^2-k^2}-\frac{y^2}{k^2-b^2}=1$ $(a>k>b)$.

Same foci: $a_E e_E = a_H e_H$; the questions on 'cuts orthogonally' with both curves centred at the origin use this.

Grade - Medium
Practice map, main concept: Sheet: J-II 81; J-M 30
Also used in: none

Asymptote

Question 72
Define asymptote for a curve.

Model Answer
A line L such that the perpendicular distance of a point P of the curve from L tends to 0 as P moves to infinity along the curve.

Grade - Low
Practice map, main concept: Sheet: C-I 5
Also used in: none

Question 73
Write equation of pair of asymptotes for standard hyperbola.

Model Answer

$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=0, \quad \text{i.e. } y=\pm\frac{b}{a}x$$

They pass through the centre, are the diagonals of the rectangle on the axes, and the axes bisect the angles between them.

Grade - Low
Practice map, main concept: Sheet: J-II 98; C-II 53
Also used in: Sheet: J-II 77; C-II 41, 50

Question 74

What is relation between equations of hyperbola, pair of asymptotes and conjugate hyperbola?

Model Answer

$$H + C = 2A$$

They differ only in the constant term: $H: \ S = 1$, $A: \ S = 0$, $C: \ S = -1$; a hyperbola and its conjugate have the same asymptotes.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: C-II 52 | DPP: 9(2)

Question 75

How to find equation of pair of asymptotes for a general hyperbola?

Model Answer

Change only the constant: $A = H + \lambda$ and choose λ so that $\Delta = 0$ (pair of lines); then $C = 2A - H$.

Asymptotes $L_1 = 0$, $L_2 = 0$ given: hyperbola $L_1 L_2 = k$ (k from a point or a tangent). Ex: $L_1 = 2x + 3y + 3$, $L_2 = 3x + 4y + 5$, through $(1, -1)$: $L_1 L_2 = 8$.

Ex: $3x^2 - 5xy - 2y^2 - 5x + 11y - 8 = 0$: λ gives asymptotes $3x^2 - 5xy - 2y^2 - 5x + 11y - \frac{38}{49} = 0$.

Grade - Low

Practice map, main concept: Sheet: C-II 52

Also used in: Sheet: C-II 46

Question 76

What is the position of a line w.r.t. hyperbola corresponding to number of common points between them?

Model Answer

$y = mx + c$: $c^2 > a^2m^2 - b^2$ two points (secant), $=$ tangent, $<$ no point. A line parallel to an asymptote ($m = \pm\frac{b}{a}$, $c \neq 0$) meets the curve in exactly one point but is not a tangent.

No tangent has $|m| \leq \frac{b}{a}$. Ex: $4x^2 - 9y^2 = 36$, slope $\frac{2}{5} < \frac{2}{3}$: no tangent.

Grade - Low

Practice map, main concept: none

Also used in: DPP: 9(5)

Question 77

How to find angle between asymptotes for a standard hyperbola?

Model Answer

$$\theta = 2\tan^{-1}\frac{b}{a}, \qquad \tan\theta = \frac{2ab}{a^2 - b^2}$$

θ = angle containing the hyperbola.

Grade - Low

Practice map, main concept: Sheet: C-II 58

Also used in: none

Question 78
What is the relation between angle between asymptotes and e?

Model Answer

$$e = \sec\frac{\theta}{2}$$

Ex: $x^2 - 3y^2 = 3$: $\frac{b}{a} = \frac{1}{\sqrt{3}}$, $\theta = 60°$, $e = \sec 30° = \frac{2}{\sqrt{3}}$. Rectangular: $\theta = 90°$, $e = \sqrt{2}$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: C-II 58

Question 79
Find number of tangents and position of corresponding point(s) of tangency from a point lying in outside region of hyperbola.

Model Answer
Between the asymptotes on the side of a branch: two tangents, both touching that branch. Between the asymptotes on the side of the conjugate axis: two tangents, one to each branch. On an asymptote (not the centre): one tangent. From the centre: none.

Two branches have no common tangent but have a common normal (the transverse axis).

Grade - Low
Practice map, main concept: Sheet: J-II 100
Also used in: none

Question 80
The tangent at P meets the asymptotes at Q and R. What can you say about P and the triangle CQR?

Model Answer
$Q\,(a(\sec\theta + \tan\theta),\, b(\sec\theta + \tan\theta))$, $R\,(a(\sec\theta - \tan\theta),\, -b(\sec\theta - \tan\theta))$: P is the mid-point of QR and area $CQR = ab$ (constant).

$CQ \cdot CR = a^2 + b^2$. Ex: $\frac{x^2}{45} - \frac{y^2}{20} = 1$ at $P(9,4)$: $Q(15,10)$, $R(3,-2)$, area $= 30 = ab$.

Grade - Low
Practice map, main concept: Sheet: J-II 77; C-II 50-51 | DPP: 10(6)
Also used in: none

Rectangular Hyperbola $xy = c^2$

Question 81
Derive everything for $xy = c^2$.

Model Answer

$$P(t) = \left(ct,\ \frac{c}{t}\right), \qquad \text{asymptotes } x = 0,\ y = 0$$

Centre $(0,0)$; TA $y = x$, CA $y = -x$; vertices $\pm(c,c)$; $e = \sqrt{2}$; foci $\pm(\sqrt{2}c, \sqrt{2}c)$; directrices $x + y = \pm\sqrt{2}c$; $\ell_{TA} = \ell_{CA} = \ell_{LR} = 2\sqrt{2}c$; aux. circle $x^2 + y^2 = 2c^2$; director circle = the point $(0,0)$.

$x^2 - y^2 = a^2$ rotated by 45° becomes $xy = \frac{a^2}{2}$. Ex: $xy = 8$: vertices $\pm(2\sqrt{2}, 2\sqrt{2})$, foci $\pm(4,4)$.

Grade - Medium
Practice map, main concept: Sheet: C-I 4; C-II 59, 61 | Question Bank: 113
Also used in: Sheet: C-II 60

Question 82
What are the tangent and normal at $P(t)$ on $xy = c^2$, and where does the normal meet the curve again?

Model Answer

$$\text{Tangent: } x + t^2 y = 2ct, \qquad \text{Normal: } t^3 x - ty = c(t^4 - 1)$$

At (x_1, y_1): tangent $xy_1 + yx_1 = 2c^2$ ($\frac{x}{x_1} + \frac{y}{y_1} = 2$). Normal at t meets the curve again at t_1 with $t^3 t_1 = -1$.

$y = mx + k$ touches $xy = c^2$ iff $k^2 + 4mc^2 = 0$. Ex: normal to $xy = 4$ at $(1, 4)$ ($t = \frac{1}{2}$) meets again at $t_1 = -8$: $\left(-16, -\frac{1}{4}\right)$.

Grade - Low

Practice map, main concept: Sheet: J-I 44-46; C-I 7 | DPP: 10(7-8, 10-11)

Also used in: none

Question 83
What is the chord of $xy = c^2$ joining t_1, t_2, and the chord with a given mid-point?

Model Answer

$$x + t_1 t_2\, y = c(t_1 + t_2), \qquad \text{mid-point } (h, k): \ \frac{x}{h} + \frac{y}{k} = 2$$

Slope of chord $= -\frac{1}{t_1 t_2}$. Mid-points of chords of slope m lie on $y + mx = 0$ (a diameter).

Grade - Medium

Practice map, main concept: Sheet: J-I 43; J-M 12; C-II 47 | DPP: 10(9, 12)

Also used in: none

Question 84
A circle meets $xy = c^2$ in four points t_1, t_2, t_3, t_4. What relations hold?

Model Answer

$$t_1 t_2 t_3 t_4 = 1, \qquad \sum x_i = -2g, \qquad \sum y_i = -2f$$

Put $(ct, \frac{c}{t})$ in $x^2 + y^2 + 2gx + 2fy + k = 0$: $c^2 t^4 + 2gct^3 + kt^2 + 2fct + c^2 = 0$. For a circle centred at the origin $\sum x_i = \sum y_i = 0$.

Grade - Low

Practice map, main concept: Sheet: J-II 76

Also used in: none